

Session Objectives

1. Establish a general approach to the vascular patient.
2. Recognize the clinical presentation of arterial, venous, and lymphatic disorders.
3. Understand the complications of arterial, venous, and lymphatic disorders.
4. Synthesize the proper workup for patients presenting with arterial, venous, and lymphatic disease.
5. Understand and appreciate therapeutic options and algorithms in the treatment of arterial, venous, and lymphatic disorders.

Approach to the Vascular Patient – History Taking

Clinical History (*timing* is very important)

- Pain, weakness in the extremities
- Neurosensory complaints (warmth, numbness, hypersensitivity)
- Discoloration, swelling, tissue loss/ulceration, varicosities

Medical History

- Hypertension
- Atherosclerotic disease (CAD, PAD, CVA, carotid stenosis)
- Hyperlipidemia
- Diabetes mellitus

Surgical History

- Everything

Social History

- Smoking

Common or Concerning Symptoms

- Abdominal, flank, or back pain
- Pain or weakness in the arms or legs
- Intermittent claudication
- Cold, numbness, pallor in the legs; hair loss
- Swelling in calves, legs, or feet
- Color change in fingertips or toes in cold weather
- Swelling with redness or tenderness

Approach to the Vascular Patient – Physical Exam

Inspection

- Meticulous inspection and documentation of skin changes
 - Atrophy, cyanosis, ulceration, hair loss, abnormal nail growth
 - Pictures are often taken to monitor wound care

Palpation of pulses

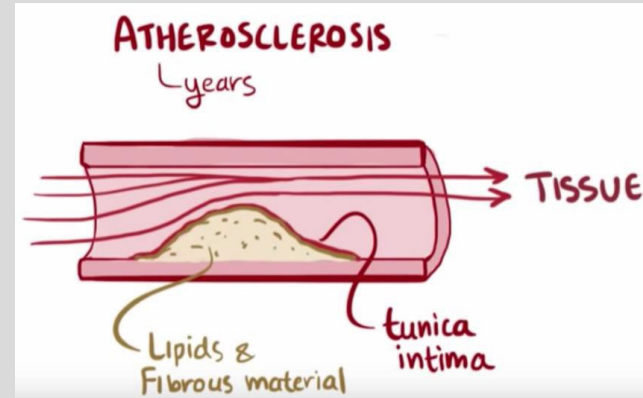
- Unilateral: Abdominal aorta
- Bilateral: carotid, brachial, radial, ulnar, femoral, popliteal, posterior tibial, & dorsalis pedis arteries

Auscultation & Ultrasound

- Auscultate for carotid & renal bruits (indicative of turbulent flow)
- Screening ultrasound for abdominal aortic aneurysm (AAA)
- Follow-up ultrasound for:
 - Patients with known or suspected vascular disease
 - Patients following a vascular intervention

Peripheral Arterial Disease (PAD)

- Obstruction or deterioration of arteries *other than those supplying the brain or the heart*. Commonly affects the lower extremity.
 - Major pathophysiologic process is **atherosclerosis**
- Patients typically present with **claudication** (pain with activity, that is relieved by rest, and reproducible with **the same level of activity**) +/- a non-healing wound
- Affects 20% of Americans ≥ 65 years old
- Increased risk with obesity, non-white ethnicities, smoking, diabetes, HTN and HLD (same risk factors as atherosclerosis)

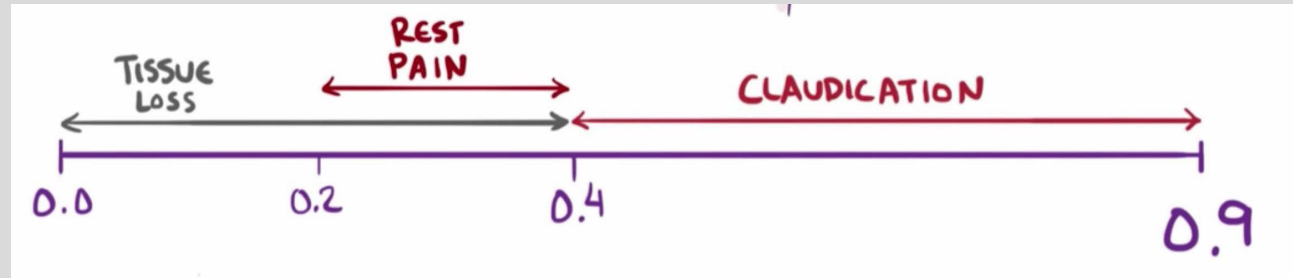




Diagnosis of PAD – Ankle-Brachial Index (ABI)

ABI of <0.9 is diagnostic for PAD

- In cases of vascular trauma or emergency, an ABI <0.9 indicates further diagnostic imaging (usually CT- or MR-angiography)
- ABI <0.4 is typically seen in patients with ischemic rest pain.
- Tissue loss occurs in patients with ABI <0.4 (these numbers vary)
- An ABI >1.3 occurs in patients with calcified arteries = non-valid!
 - Tends to occur in diabetics with ESRD



PAD – Complications

Acute Limb Ischemia (ALI)

Acute thromboembolic occlusion of the extremity, typically from **cardiac origin**, that most often lodges at a site of a **pre-existing plaque**.

Clinical Presentation (6 P's)

Pulselessness

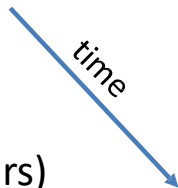
Pallor

Pain

Paresthesia

Paralysis (~6 hours)

Poikilothermia (coolness)



Chronic Limb-Threatening Ischemia (CLTI)

1-2% of patients with PAD

Clinical Presentation

Rest pain (ABI <0.4)

Tissue loss (ulcers, gangrene etc...)

1-Year Prognosis

- 50% alive with both limbs
- 25% have amputation
- 25% CV-related mortality





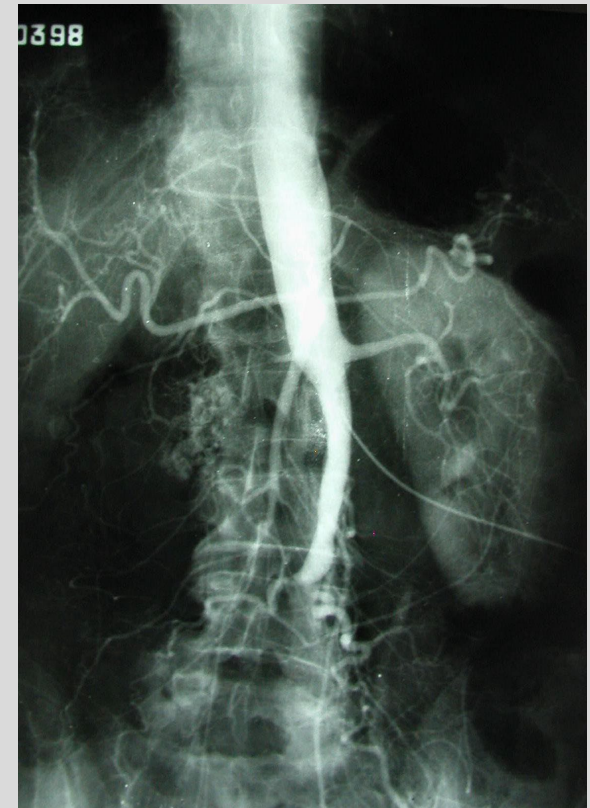
PAD – Leriche Syndrome

Aortoiliac Occlusive Disease – Affects the distal aorta, iliac arteries +/- femoropopliteal vessels

Clinical Presentation – Triad

- Buttock & thigh claudication
- Decreased femoral pulses
- Erectile dysfunction

Always keep **Leriche syndrome** on your differential diagnosis of erectile dysfunction and hip/thigh pain!



PAD – Treatment Options

Conservative Treatment (ABI>0.4, no CLTI [rest pain, tissue loss])

- Lifestyle changes: Smoking cessation, exercise, weight loss
- Medication: Statin, ASA, cilostazol (PDE3 inhibitor), blood glucose & BP control

Note: Vasodilators do not halt disease progression, just symptom relief

PAD patients are more likely to die of an MI or stroke than lose a limb!

>70% stenosis and/or **presence of CLTI** are *general* indications for open surgery (i.e. bypass of affected artery) or endovascular intervention (e.g. balloon dilation, stenting, atherectomy). Medical treatment is continued, along with constant wound care.

ALI is treated aggressively with anticoagulation (heparin), and subsequent imaging determines which intervention can be utilized.

Thrombolytics (tPA) + mechanical thrombectomy often used.

Question Break

A 68-year-old female presents to the clinic because of increased right upper calf pain when walking. She used to be able to walk a mile to work each morning but recently has noticed that at the half-way point, she must stop for a few minutes to “let her legs rest”. This happens every morning at the same distance from her house. She has a PMHx of HTN, DM and had a STEMI 3 years ago where she had a coronary stent placed in her LAD. She has been smoking a pack per day for the last 30 years and has no intention of quitting. She is currently taking Plavix, simvastatin, lisinopril and metformin. Physical exam shows a non-healing wound on her right lower leg.

What is the first step in the diagnosis of this patient?

At what artery is her blockage most likely?

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What is the first step in the diagnosis of this patient? Meticulous physical exam and perform an ABI

At what artery is her blockage most likely? Popliteal artery

Question Break

A 72-year-old female presents to the emergency department after 3 hours of excruciating left lower extremity pain. She reports no inciting events or recent trauma. Past medical history is relevant for coronary artery disease, treated with a 4-vessel CABG in 2012, atrial fibrillation, type-2 diabetes mellitus, hyperlipidemia, and COPD. On physical exam, you note her left leg is paler in color and cooler to the touch, when compared to the right. She has +2 femoral, popliteal, posterior tibial, and dorsalis pedis pulses on the right, but you are only able to appreciate a +2 femoral pulse on the left.

What is the most likely etiology of this patient's current condition?



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Atherosclerosis



Carotid Stenosis – Anatomy & Etiology

Transient Ischemic Attack (TIA)

- Acute (<24 hr) loss of focal cerebral function
 - Focal = correlates to one vascular supply of the brain (typically MCA).
Presents with localized symptoms
- Most are brief, lasting 2-15 minutes and rapid in onset
- Typically unilateral motor & sensory loss, aphasia, or dysarthria

Transient Monocular Blindness (TMB) aka amaurosis fugax

- Transient loss of unilateral vision due to ophthalmic a. occlusion
- “A shade came over my eye”
 - Location of vision loss correlates with occlusion of retinal artery branch
- Fundoscopy: Hollenhorst plaque (carotid atheroemboli)

Physical Examination

- Carotid bruit
- Screening test = Duplex ultrasound





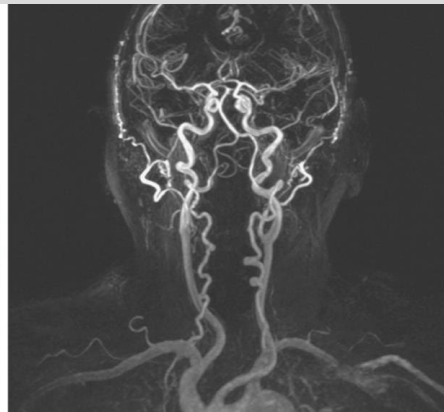
Carotid Stenosis – Diagnosis

Routine screening & first imaging modality of choice = **Duplex U/S**

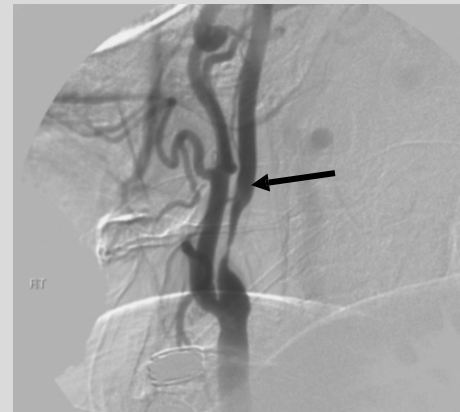
- Duplex only provides an **estimate** of stenosis based on **PSV**
- Follow-up tests include MRA or CTA
 - These modalities can also image vertebrobasilar & cerebral arteries
- Gold standard remains DSA, but is rarely done for diagnosis
 - Carotid stenting utilizes DSA intra-operatively for placement



CTA w/ 3D reconstruction



MRA w/ cerebral vessels



Digital subtraction angiography



Carotid Stenosis – Treatment Options

Treatment is determined based on presence of symptoms and degree of stenosis. In general, **any stenosis indicates need for antiplatelet (ASA +/- clopidogrel) and statin therapy**. Indications for surgical procedures is not important to know at this point.

Carotid Endarterectomy (CEA)

- Open surgery with great results, where carotid artery is entered via incision (arteriotomy), and plaque is removed (atherectomy)
- Risk of cranial nerve injury (CNs IX, X, XI, & XII) & stroke

Carotid Stenting (CAS)/Trans-Carotid Revascularization (TCAR)

- Minimally invasive techniques for patients unable to tolerate open surgery
- Stent is placed via femoral access (CAS) or carotid access (TCAR)
- CAS = controversial due to poor outcomes when compared to CEA
- TCAR = new “hybrid” technique involving localized stent placement

Landmark clinical trials for CEA vs CAS: CREST & CREST-2 (ongoing)

CVI – Pathophysiology

CVI can be primary or secondary in nature

- **Primary** → due to anatomic defects in veins or valves
- **Secondary** → post-thrombotic syndrome (PTS) following DVT, leads to venous occlusion and inflammation that damages lumens & valves

Natural History of CVI



Varicose veins
Valves cannot handle elevated pressure → reflux



→ **Edema**
Extravasation of RBCs into the interstitium



→ **Pigmentation**
Skin breakdown → infection of exposed tissue or blood within subQ space



→ **Ulceration**
Mechanical wall stress → release of inflammatory mediators

Venous ulcers are most commonly found 2-3 cm above medial malleolus

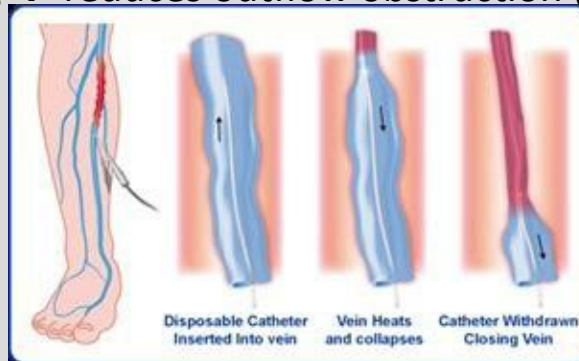
CVI – Treatment

Conservative Therapy

- Compression stockings → external pressure assists venous return
- NSAIDs for pain relief

Interventions

- Sclerotherapy → injected chemicals or foam to shrink varicosities
- Endovenous ablation → laser or radiofrequency signals create heat, which closes vein → fibrosis leads to permanent vein closure
 - Venous blood is then re-routed to more competent & deeper veins
- Iliac vein stenting → reduces outflow obstruction (i.e. MTS)



Endovenous Ablation



Sclerotherapy

Question Break

A 38-year-old woman presents to your attending vascular surgeon's clinic and you are the first to see her. She complains of "gross veins popping out of her leg", swelling, and pain while standing at her job as a barista. The pain and edema worsen as the day progresses. On physical exam you note an overweight (BMI = 29.6) female with multiple, 2- and 3-cm varicose veins of the left and right lower extremity that are compressible and non-tender.

Duplex ultrasonography reveals compression of left common iliac vein.

What is the most likely etiology of this venous compression?

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May-Thurner Syndrome

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Arterial vs Venous Ulcers – Comparison

Characteristic	Arterial Ulcer	Venous Ulcer
History	Smoking, PAD, CAD, diabetes, previous vascular disease.	DVT, varicose veins, obesity, trauma, surgery, pregnancy.
Location	Distal extremities (i.e. toes, fingers) & bony prominences.	Between lower calf and medial malleolus.
Ulcer Bed Ulcer Borders	Dry, necrotic tissue. Regular, punched-out margins.	Fibrous material with exudate. Irregular margins.
Condition of Extremity & Pulse Exam	Thin, shiny, hairless skin. Pallor, reduced temperature. Diminished pulses & capillary refill.	Brawny, thick, itchy & fibrotic skin. Pigmentation with edema. Normal pulse & capillary refill.
Treatment	Revascularization, anti-platelets, statins, & risk-factor modification (e.g. smoking, blood-glucose).	Compression therapy, leg elevation, endovenous ablation (especially of perforator veins), iliac vein stenting.



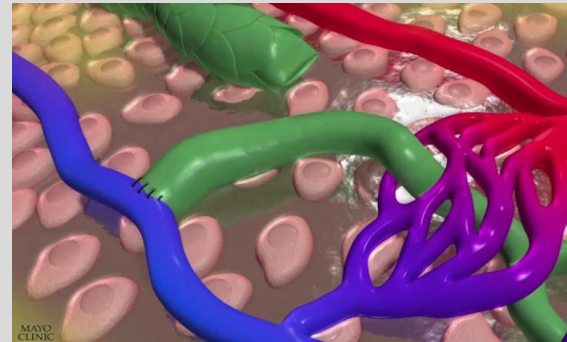
Lymphedema – Management & Treatment

Conservative Therapies

- Compression or velcro stockings
- Exercise
- Pneumatic compression devices
- Lymphatic OMT
- **Gold standard** = manual decompressive therapy

Surgical Interventions

- Lymphovenous bypass
- Vascularized lymph node transfer
- Excisional debulking
- Liposuction → remember, lymphedema contains a lot of adipose tissue!



Lymphovenous bypass

Question Break



While on your internal medicine rotation at Queens Hospital Center, a 56-year-old man who recently arrived from Bangladesh is admitted to your team's service. He complains of progressive limb swelling over the past 3 years. He works as a farmer and has not seen a doctor in 2 decades. He reports a past medical history of diabetes and high cholesterol. He smokes 1.5 packs of cigarettes per day, and has been smoking since he was a teenager. On physical exam you notice a markedly enlarged right lower limb that is firm and tough to the touch. Elevation of the leg does not relieve the swelling. A blood smear is collected and Giemsa staining reveals a long, cylindrical worm.

What is the most likely inciting factor of his disease process?

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What is the most likely inciting factor of his disease process? *W. bancrofti*

Overview

Approach to the Vascular Patient

- History, physical (inspect, palpate, auscultate, & ultrasound)
- Meticulously document all physical exam findings and pulse exam (pictures included)
- Imaging: U/S, CT- & MR-angiography, digital subtraction angiography (DSA)

Arterial Disease

- Peripheral Arterial Disease → atherosclerosis of lower extremity
 - Claudication: pain with activity, relieves with rest, reproducible
 - Leriche syndrome, aortoiliac occlusion: claudication, decreased pulses, & erectile dysfxn
 - ALL: acute thromboembolic occlusion at previously stenotic site
 - 6 P's: pain, pallor, pulseless, paresthesia, paralysis, poikilothermia
 - CLTI: rest pain, tissue loss with high-risk of morbidity & mortality
 - ABI = fraction of highest ankle pressure / highest brachial pressure
 - <0.9 = PAD; <0.4 = rest pain; <0.4 = tissue loss
- Carotid Stenosis → atherosclerosis of bifurcation; leading cause of stroke
 - Presents with contralateral TIA and/or ipsilateral amaurosis fugax
 - Screen with duplex U/S, follow-up with CTA or MRA